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[date]

Guest Editors, Special Issue “Artificial Intelligence and Computational Mechanics for Materials Design”

Extreme Mechanics Letters

Dear Dr. Tac, Dr. Moreno-Mateos, Dr. Hossain and Prof. Zhang,

On behalf of all co-authors, I submit the manuscript “[**title**]” by K. Nishioka, F. Klempt, H. Geisler, R. Mukherjee, N. Heine, K. Doll-Nikutta, M. Stiesch, S. P. Szafranski, M. Soleimani and P. Junker for the special issue “Artificial Intelligence and Computational Mechanics for Materials Design” (VSI: AI & CM), following the invitation sent to Prof. Junker.

Biofilms are living materials whose composition, and with it their mechanical behaviour, is set by interactions between species that cannot be measured directly. We identify these interactions as a physics-constrained inverse characterization problem: a thermodynamically consistent multiphase model, derived from the extended Hamilton principle, fixes the structure, and Bayesian inference determines its 15 interaction coefficients from in-vitro time courses of a five-species oral biofilm. The forward model is compiled in JAX and evaluated for all particles in parallel on a GPU, which makes it affordable to repeat every inference stage with independent seeds and to accept a stage only after explicit checks of mixing, seed agreement and posterior mass at the prior bounds. **[Two sentences on the main result: which interactions the data determine, and the in-silico knockout prediction.]**

We believe the work fits the special issue because it combines a mechanistic, variationally derived continuum model with data-driven probabilistic inference for the inverse characterization of a soft biological material, and because it makes explicit which conclusions the data support—the kind of physics-based guidance that the call identifies as essential for data-driven methods in materials science.

The manuscript has not been published and is not under consideration elsewhere. All authors have approved the submission [**confirm**]. Code, run configurations, convergence records and posterior samples are archived at Zenodo (**DOI**). The authors declare no competing interests.

Sincerely,

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